

Report

Motion Detection Alarm System

Introduction to Embedded Systems

An embedded system is a combination of hardware and software designed to perform a specific task. Unlike general-purpose computers, embedded systems are developed for dedicated applications such as automation, monitoring, and control systems.

Embedded systems are widely used in:

- Smart home devices
- Automotive systems
- Healthcare devices
- Industrial automation
- Robotics
- Security systems

Examples of embedded systems include washing machines, traffic light controllers, smart watches, home automation systems, and motion detection alarms.

Difference Between Microcontroller and Microprocessor

Microcontroller

Designed for specific tasks

Contains CPU, memory, and I/O ports in a single chip

Low power consumption

Used in embedded systems

Cost-effective

Microprocessor

Designed for general-purpose computing

Requires external memory and peripherals

Higher power consumption

Used in computers and laptops

More expensive

Component Study

1. Arduino UNO

Arduino UNO is a microcontroller board based on the ATmega328P microcontroller. It is commonly used in embedded systems and IoT projects because of its simplicity and ease of programming.

Functions of Arduino UNO:

- Reads signals from sensors
- Processes data
- Controls output devices such as LEDs and buzzers
- Executes programmed instructions

2. PIR Motion Sensor

PIR (Passive Infrared) Sensor is used to detect human motion. It senses infrared radiation emitted by the human body.

Working of PIR Sensor:

- Detects changes in infrared radiation
- Sends a signal to the Arduino when motion is detected
- Used in security and automation systems

3. LED

LED (Light Emitting Diode) is used as a visual indicator in the project.

4. Buzzer

The buzzer produces sound when motion is detected and acts as an alarm.

5. Breadboard and Jumper Wires

Breadboard and jumper wires are used for making temporary circuit connections without soldering.

Project Title: Motion Detection Alarm System

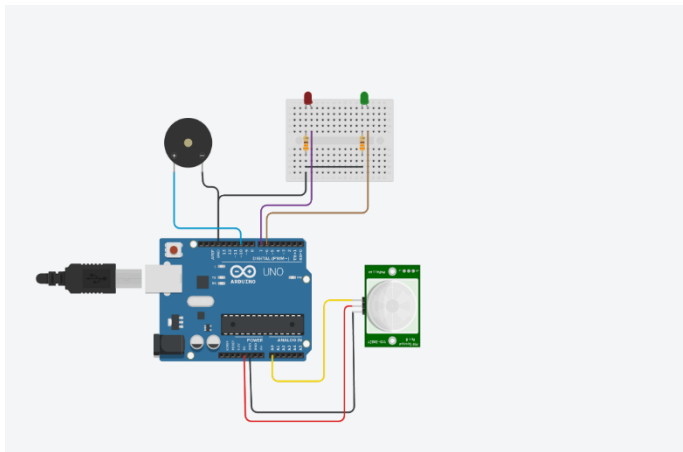
Objective of the Project:

The objective of this project is to detect human movement using a PIR motion sensor and generate an alert using an LED and buzzer.

This project demonstrates the practical implementation of embedded systems and IoT-based security applications.

Circuit Diagram

The circuit for the Motion Detection Alarm System was designed and simulated using Tinkercad Circuits.



Working Principle

The Motion Detection Alarm System works using a PIR motion sensor connected to an Arduino UNO.

Step-by-step working:

1. The PIR sensor continuously monitors the surrounding area for infrared radiation.
2. Human bodies emit infrared radiation.
3. When a person moves near the sensor, the infrared pattern changes.
4. The PIR sensor sends a signal to the Arduino UNO.
5. The Arduino processes the signal.
6. If motion is detected:

- The LED turns ON.
 - The buzzer produces sound.
7. When no motion is detected:
- The LED remains OFF.
 - The buzzer remains OFF.

Code Explanation

- The PIR sensor is connected to analog pin A0.
- LEDs are connected to digital pins 6 and 7.
- The buzzer is connected to pin 10.
- The Arduino continuously reads sensor values.
- If motion is detected, the buzzer and LED are activated.
- If no motion is detected, the system remains idle.

Applications of the Project

Applications of the Motion Detection Alarm System include:

- Home security systems
- Intruder detection systems
- Smart surveillance systems
- Automatic warning systems
- Office and shop security
- Smart home automation

Advantages of the Project

- Simple and low-cost system

- Easy to implement
- Useful for security purposes
- Low power consumption
- Real-time motion detection

Tools and Software Used

- Tinkercad Circuits
- Arduino IDE
- Arduino UNO

Conclusion

The Motion Detection Alarm System successfully detects movement using a PIR motion sensor and alerts users through an LED and buzzer.

This project helped in understanding:

- Basics of embedded systems
- Sensor interfacing
- Arduino programming
- IoT-based security applications

The project demonstrates how embedded systems can be used for automation and smart security solutions.

References

- Arduino Official Documentation
- Tinkercad Circuits
- Embedded Systems Learning Resources